

Nan Yan | CS Ph.D. Applicant

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Education

Wuhan University

GPA: 90.07/100, National Scholarship (Top 0.2%), Advisor: Prof. [Li Yuqing](#)

School of CSE

M.Eng. 2023–Now

Shandong University

GPA: 88.51/100, Graduated with Outstanding Honor

School of CST

B.Eng. 2019–2023

Research Interests

Trustworthy AI (LLMs and Agent Security), Privacy-Preserving Machine Learning (FL, HE, DP)

Academic Experience

Publications

EmbedX: Embedding-Based Cross-Trigger Backdoor Attack Against Large Language Models

USENIX Security Symposium (acceptance rate: 17.1%)

2025

Nan Yan, Yuqing Li, Xiong Wang, Jing Chen, Kun He, Bo Li

FedPHE: A Secure and Efficient Federated Learning via Packed Homomorphic Encryption

IEEE Transactions on Dependable and Secure Computing (TDSC)

2025

Yuqing Li ([advisor](#)), **Nan Yan**, Jing Chen, Xiong Wang, Jianan Hong, Kun He, Wei Wang, Bo Li

Efficient and Straggler-Resistant Homomorphic Encryption for Heterogeneous Federated Learning

IEEE International Conference on Computer Communications (INFOCOM, acceptance rate: 19.6%)

2024

Nan Yan, Yuqing Li, Jing Chen, Xiong Wang, Jianan Hong, Kun He, Wei Wang

Manuscripts (Under Review)

MemIncept: Steering LLM Agents via Cooperative Stealthy Memory Injections

Nan Yan, Qian Lou, Jiarong Xing

2026

Efficient and Differentially Private Federated LLM Fine-Tuning on Heterogeneous Clients

Nan Yan, Yuqing Li, Jing Chen, Xiong Wang, Wei Wang, Kun He, Ruiying Du, Shuhua Li

2026

CoMoMark: Cross-Modal Collaborative Backdoor Watermarking for Vision-Language Models

Huiyi Tang, Yuqing Li, Yushi Yang, Xiong Wang, Haoran Wang, **Nan Yan**, Kun He, Jing Chen

2026

Federated LoRA Fine-Tuning with Pipelined Error-Mitigated Aggregation and Matrix-Wise Freezing

Haoran Wang, Xiong Wang, Yuqing Li, Jing Chen, Junyi Zhang, **Nan Yan**, Kun He, Wei Wang

2026

FedP2P: Towards Accelerated Federated Learning with Peer-to-Peer Communication

Haoran Wang, Xiong Wang, Yuqing Li, Jing Chen, Yuntao Nie, **Nan Yan**, Meng Jin, Bo Li

2025

Selected Projects

LLM and Agent Security

MemIncept: Bidirectional Set Evolution for Agent Memory Injection Attack

2025–2026

- Built a bidirectional evolutionary attack that injects cooperative benign queries into LLM-agent memory, optimizing retrievability and reasoning bias to achieve highly stealthy and effective agent manipulation.

- Rice University, working with Prof. [Jiarong Xing](#) and submitted to ICML 2026.

EmbedX: Efficient and Stealthy Cross-Trigger Backdoor Attack on LLMs

2024–2025

- Designed an embedding-level soft-trigger mechanism with latent frequency–gradient constraints, enabling multi-trigger activation with high stealthiness and robustness against fine-tuning.

- Delivered 100% ASR while reducing trigger-switching overhead from 4000s to 0.5s and lowering false-trigger rate to 1%, outperforming SOTA backdoor baselines on 5 datasets & 4 LLMs.

- 4.6 kloc in [codebase](#), and accepted in the Proc. of [USENIX Security 2025](#).

Privacy-Preserving Federated Learning

FedPHE: Efficient Homomorphic Encryption for Heterogeneous Federated Learning

2023–2024

- Proposed a contribution-aware encrypted weighted aggregation and sketching-based client selection algorithm for optimizing FL training efficiency with CKKS-based HE.
- Achieved a training speedup of 1.85–4.44×, cut the communication overhead by 1.24–22.62×, and reduce the straggler effect by up to 1.71–2.39×, outperforming 6 baselines on 4 datasets.
- 4.5 kloc in [codebase](#), work earned 39 stars and accepted in the Proc. of [IEEE INFOCOM 2024](#).

Federated Learning with Knowledge Distillation

2022–2023

- Designed and implemented a federated learning framework with knowledge distillation and 5 baseline methods, earning 26 stars in [codebase](#) for facilitating efficient experimentation across diverse datasets and models.

Open Source Software

Pioneer: a low-cost Bluetooth-based virus tracking system

2022–2024

- Built a cost-effective virus tracking system leveraging Bluetooth technology, secured with the national cryptographic algorithm SM3 for encryption and certification. 19.6 kloc, [codebase](#) released and work earned 153 stars, 85 forks.

Awards

Research Fundings

International Conference Support: Wuhan University

2025

Travel Support: USENIX Security’25 Student Grant

2025

Cyberspace Security Innovation Grant: TOPSEC’23 & Huawei’25

2023, 2025

Scholarship

National Scholarship (**Top 0.2%**): Ministry of Education, China

2024, 2025

BYD Scholarship (**Top 3**): Dept. CSE, Wuhan University

2025

First Class Scholarship (**Top 10%**): Wuhan University

2024, 2025

Second Class Scholarship (**Top 10%**): Shandong University

2020, 2022

Merit-Based Scholarship: Shandong University

2020, 2022×4

Honors

Metrit Student (**Ranking: Top 1%**): Wuhan University

2024, 2025

Excellent Student Cadre: Shandong University’22 & Wuhan University’25

2022, 2025

Outstanding Graduate Award (**Top 15%**): Shandong University

2023

Alumni Council Representative (**Top 6/101**): Dept. CST, Shandong University

2023

The Power of Role Models (**Top 1**): Dept. CST, Shandong University

2023

Competitions

Second Prize: National Open Source Award Program, China

2025

Second Prize: National College Cryptography Mathematics Contest, China

2022

Second Prize: National College Student Information Security Competition, China

2022

First Prize: National College Cryptography Mathematics Contest, North China Division

2022

First Prize: China Undergraduate Mathematical Contest in Modeling, Shandong Province

2021

Patents and Software Copyrights

Differential privacy-based heterogeneous federal fine tuning language model method and system

China Patent Application ZL 2024 1 1379992.3, PatentGrant

Sep 2025

Cross-silo heterogeneous federated learning system based on homomorphic encryption V1.0

Software Copyrights License 2024SR1516588

Oct 2024

Technical Skills

Languages: Mandarin Chinese(native) and English (proficiency, IELTS: 7.0)

Technical Expertise: Agent Systems, LLM Fine-tuning, Adversarial Attacks & Defenses